

Health Effects of Air Pollution Exposure on Children and Adolescents in São Paulo, Brazil

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Summary. Children and adolescents have been considered more susceptible to the effects of air pollution than adults. In order to investigate the responses of children of different ages to air pollution exposure, daily records of hospital admissions for children in five age groups (equal or less than 2 years of age, 3–5, 6–13, 14–19, and all ages together, i.e., from 0–19 years of age) were obtained from January 1993 to November 1997 in São Paulo, Brazil, and were compared to daily records of PM₁₀, O₃, SO₂, CO and NO₂ concentrations in ambient air. For each age group a generalized additive Poisson regression was fitted controlling for smooth functions of time, temperature, humidity, and days of the week, with an additional indicator for holidays. Polynomial distributed lag models were used to estimate the 7-day cumulative effect of each pollutant.

Children 2 years or less were the most susceptible to the effects of all five pollutants with an increase of 9.4% (95% CI: 7.9,10.9) in respiratory admissions associated with each interquartile range increase in PM₁₀. The oldest group was the second most susceptible to air pollutants, with each interquartile range increase in PM₁₀ associated with a 5.1% (95% CI: 0.3,9.8) increase in respiratory admissions. An interquartile range increase in CO was associated with an 11.3% (95% CI: 5.9,16.8) increase in respiratory hospitalizations. When a multipollutant model was used, the effect of PM₁₀ on respiratory admissions for all ages together was unchanged, while the SO₂ and the other pollutants effect was substantially reduced.

This study showed that daily respiratory hospital admissions for children and adolescents in São Paulo increased with air pollution, and that the largest effects were found for the youngest (2 years or less) and oldest (14–19 years) age groups. **Pediatr Pulmonol.** 2001; 31:106–113.

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Key words: air pollution; respiratory diseases; hospital admissions; children; adolescents; health effects.

INTRODUCTION

The susceptibility of children and elderly adults to air pollutants exposure has been reported by many authors in both mortality^{1–4} and morbidity^{5,6} studies. Air pollution has been associated with general and respiratory post-neonatal infant mortality in developing^{3,7–10} and developed¹¹ countries. Morbidity studies have similarly demonstrated deleterious air pollution effects using respiratory symptoms,^{12–16} lung function,^{17,18} emergency room visits,^{19,20} and hospital admissions^{21,22} as outcomes. Recent studies have suggested that air pollution is associated with prenatal events. Pereira et al.²³ showed variations in late fetal deaths positively associated with air pollution in São Paulo. Dejmek et al.²⁴ reported that maternal exposure to particulate matter was associated with intrauterine growth retardation.

Elderly age groups are reported to present different responses to environmental exposure.²⁵ Children and adolescents are likely to be at least as heterogeneous in their response to air pollution as the elderly. The accelerated process of growth and development in childhood implies significant differences in organ and

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system maturity and differences in the response of these systems to external insults. It is plausible that in response to the same adverse condition, such as air pollution exposure, different age groups may respond differently. Children younger than 2 years old seem to be more susceptible to the air pollution effects.^{9,10} It is also plausible that different responses will be exhibited to different air pollutants.

In a previous study of São Paulo children,²² an overall estimate of the association between respiratory hospital admissions and air pollution was assessed for children from 1 month of life to adolescents younger than 13 years of age. On average, 68 admissions per day were registered in hospitals supported by the Public Health System in São Paulo between October 1993 and October 1994. This number was large enough to assess health effects in the combined cohort of children. In the present study, this record was extended to 5 years of pediatric and adolescent respiratory hospital admissions in São Paulo's Public Health System. This larger sample allowed assessment of air pollution effects as a function of age.

DATA

Records of daily admissions for children and adolescents younger than 20 years old to 114 hospitals receiving public financial support were obtained from the Department of Data Analysis of the Brazilian Public Health System (DATA-SUS) for the period January 1993–November 1997. In the 5 years analyzed in this study, approximately 50% of São Paulo's inhabitants were provided with care by the Public Health System.²⁶ Respiratory admissions (ICD-9: 460–519) were divided into four age groups: equal or less than 2 years of age, 3–5, 6–13, and 14–19. An additional group, i.e., all ages together (from 0–19 years of age), was also analyzed.

ABBREVIATIONS

CI	Confidence interval
CO	Carbon monoxide
CETESB	São Paulo State Sanitary Agency
DATA-SUS	Department of Data Analysis, Brazilian Public Health System
ICD-9	International Classification of Diseases, 9th Revision
IQR	Interquartile range
Loess	Locally weighted regression smoother
$\mu\text{g}/\text{m}^3$	Micrograms per cubic meter (of air)
NO_2	Nitrogen dioxide
O_3	Ozone
ppm	Parts per million (volume)
PM_{10}	Particulate matter with diameters of 10 μm or less
RR	Relative risk
SD	Standard deviation
SEADE	State Data Analysis System Foundation (São Paulo, Brazil)
SO_2	Sulfur dioxide
Span	Smoothing parameter

Daily records of PM_{10} , O_3 , SO_2 , CO , and NO_2 concentrations in ambient air were obtained from the São Paulo State Sanitary Agency (CETESB).^{27–31} Twenty-four-hour averages were reported by CETESB for the period starting at 4:00 PM of the preceding day. Due to an upgrade of the monitoring system in July 1996, the number of pollutant monitors changed from 12 to 13 for PM_{10} , from 13 to 6 for SO_2 , and from 5 to 6 for NO_2 . The number of stations that measured the highest 8-h moving average of CO and the highest hourly average of O_3 changed from 5 to 7 and from 4 to 7, respectively. The Institute of Astronomy and Geophysics of the University of São Paulo provided daily records of minimum temperature of the day and relative humidity at noon.

METHODS

Generalized additive Poisson regressions³² were fit to the logarithm of the expected value of hospital admissions in each age group as the sum of smooth and linear functions of the predictor variables. A smooth function is a non-parametric tool that allows control for potential nonlinear dependency of the variable of interest (respiratory hospital admissions) on covariates such as season and weather. The non-parametric function adopted was loess,³³ i.e., a moving regression smoother. This approach has been used frequently in air pollution time series.³⁴

A smoothing parameter (span) must be chosen for each covariate. Age-specific smoothed parameters for time were chosen to remove long-term trends from the data, removing seasonality, and minimizing the autocorrelation of the residuals. Each admission is an independent event, and autocorrelation in residuals indicates there are omitted time-dependent covariates whose variation may confound the air pollution associations. If the autocorrelation is removed, the remaining variation from omitted covariates has no systematic temporal pattern, and hence confounding is less likely. When necessary, autoregressive terms were incorporated to eliminate serial correlation from the residuals.

Smoothing parameters were defined for temperature (2-day moving average), relative humidity (2-day moving average), and day of the week in each group, with spans chosen to minimize Akaike's information criterion.³⁵ An indicator for holidays was also included in the models.

In order to reduce sensitivity to outliers in the dependent variable, we used robust regression (M-estimation). S-Plus[®] version 4.5³⁶ was used to carry out the statistical analyses.

Distributed Lag Model

Air pollution can induce health effects not only on the day of the exposure but also on subsequent days. Hence the number of admissions on a given day will depend on the same-day effect of that day's pollution, but also on the

lagged effect of the previous day's pollution. Different approaches have been tried in order to estimate the contribution of each day of exposure on the endpoint. In our previous study,²² we observed a lagged association ranging from the day of the event to 6 days prior; therefore, associations were estimated using lagged moving averages.

In an unrestricted distributed lag model we have:

$$\text{Log}(E(Y)) = \alpha + \text{covariates} + \beta_0 X_t + \dots + \beta_q X_{t-q} + \varepsilon_t \quad (1)$$

where Y is the daily count of admissions, $E(Y)$ is the expected value of that count, and X_{t-q} is the pollutant concentration q days before the admissions. In this analysis we adopted a constrained lag structure, i.e., a polynomial distributed lag model. Usually applied to social sciences studies and econometrics,³⁷ the use of this approach in epidemiology has been described in air pollution time series.^{38,39}

In this study, the 7-day cumulative effects (lag 0 through lag 6) were estimated and we applied a sixth degree polynomial, which imposes minimal constraints, giving enough flexibility to estimate a biologically plausible lag structure, and controlling multicollinearity better than an unrestricted lag model. The sum of the effect of a unit increase in air pollution on a single day is the sum of $\beta_0 + \dots + \beta_6$. The standard errors of the estimates are given by the variance of the sum of $\beta_0 + \dots + \beta_6$, adjusting for overdispersion. The 95% confidence intervals were estimated assuming normality of the residuals.

Multipollutant Model

In order to explore the behavior of the pollutants in a multipollutant model, pollutant-specific lagged moving

averages were included. Models using up to 7-day moving averages for each pollutant were tested, and the moving averages that exhibited the biggest coefficient were included in the multipollutant model.

RESULTS

Table 1 presents descriptive analyses for the variables in the study. Except for SO_2 , all other pollutants exceeded their respective air quality standard, although declining trends were apparent during the period.

Between November 4, 1994 and May 9, 1996, no NO_2 measures were made. Therefore, for the polynomial distributed lag models, only the period without missing values were considered (from May 10, 1996 to November 30, 1997). Data were more complete for the other pollutants. Missing values were replaced with the mean of the prior and following days. Due to the difference in time-series length, specific modeling was done to evaluate the effects of NO_2 .

Admissions of children 2 years or less were more frequent than other age groups. Pneumonia and bronchopneumonia (ICD-9: 480–487) were the main cause of hospital admissions (71%) in the youngest group (Fig. 1), but decreased to 30–40% of admissions in the other groups. Respiratory admissions due to acute bronchitis, bronchiolitis, and asthma (ICD-9: 466 or 493) were more important in the intermediary age groups, reaching around 20%. There was a small fraction of upper respiratory tract acute infections (ICD-9: 460–465) in all age groups. However, the largest increase by age was among respiratory admissions not included in these categories. Chronic diseases of tonsils and adenoids are the main cause of admissions in this category.

The air pollutants exhibited a high degree of interdependence. Table 2 shows Pearson correlation coefficient.

TABLE 1—Descriptive Analyses of Variables in the Study¹

Variables	Daily mean	SD	Minimum	IQR	Maximum	N
RESP/age groups (counts of events)						
≤ 2 years	46	19.0	7	25	114	1,795
3–5 years	10	4.4	0	6	28	1,795
6–13 years	10	4.7	0	7	27	1,795
14–19 years	4	2.4	0	3	17	1,795
All ages	69	24.4	16	35	157	1,795
Pollutants						
PM_{10} ($\mu\text{g}/\text{m}^3$)	66.5	30.4	16.0	35.0	230.5	1,746
O_3 ($\mu\text{g}/\text{m}^3$)	65.0	37.4	3.9	45.8	268.5	1,471
SO_2 ($\mu\text{g}/\text{m}^3$)	21.4	11.2	1.6	14.4	76.1	1,687
CO (ppm)	4.8	2.3	0.6	2.9	19.1	1,738
NO_2 ($\mu\text{g}/\text{m}^3$)	141.4	71.2	25.0	80.5	652.1	893
Weather						
Temperature ($^{\circ}\text{C}$)	15.2	3.4	0.6	5.3	22.4	1,795
Humidity (%)	80.7	9.0	42.9	10.7	97.6	1,795

¹SD, standard deviation; IQR, interquartile range; N, number of days of observation; Temperature, minimum temperature of the day; RESP, respiratory disease-related hospital admissions.

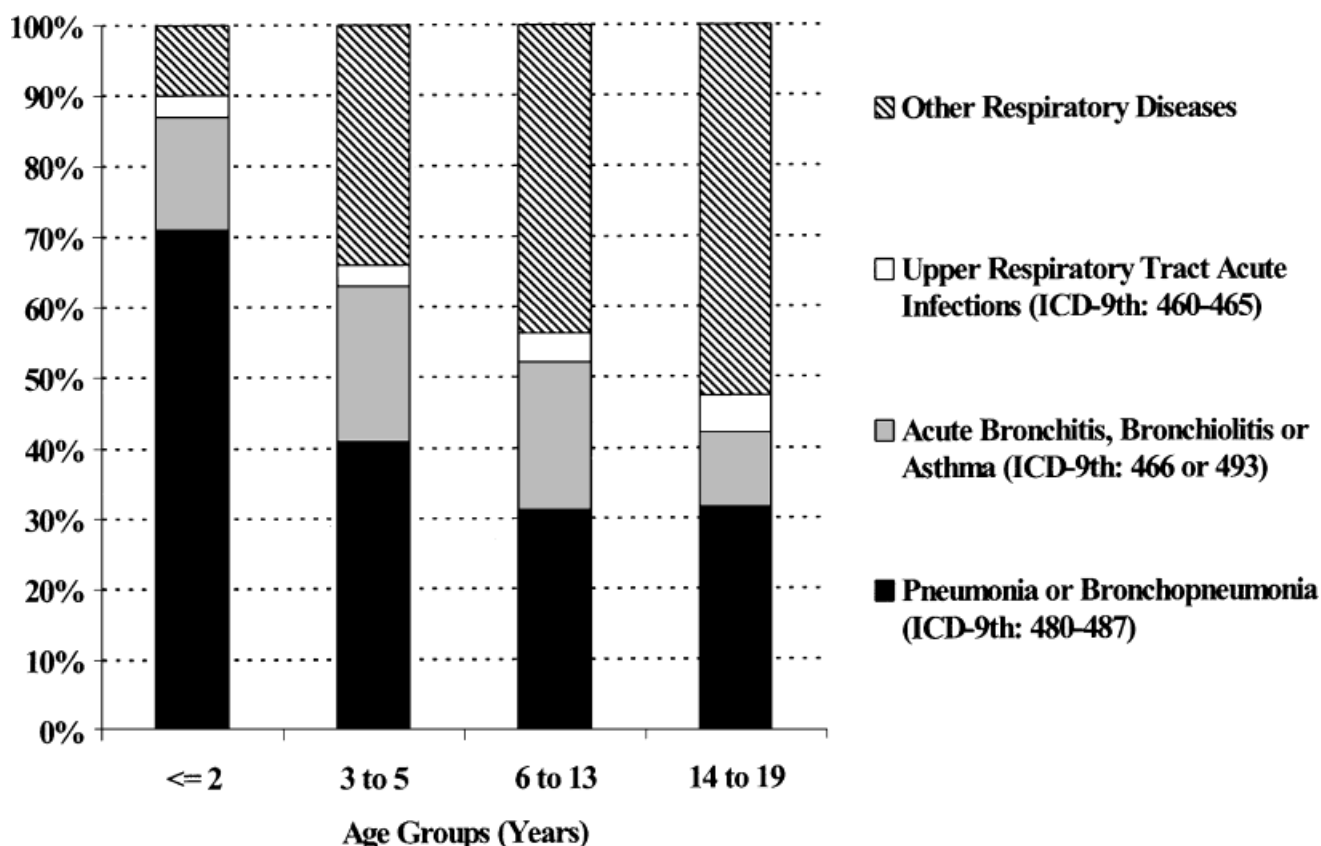


Fig. 1. Percentage of respiratory admissions (ICD-9th: 460–519) due to pneumonia or bronchopneumonia (ICD-9th: 480–487), acute bronchitis, bronchiolitis or asthma (ICD-9th: 466 or 493), upper respiratory airways acute infections (ICD-9th: 460–465), and other respiratory diseases not included in any of the three groups, according to age categories.

cients between pollutants and weather variables. There was a high correlation between traffic-related pollutants (CO and NO₂) and between these pollutants and PM₁₀.

Table 3 presents the smoothing parameters for the covariates included in the models. The wide range of spans demonstrates the necessity of individualized adjustment for each age group to control seasonality. The dependence of admissions on both temperature and humidity was almost linear (spans close to 1). For days of the week,

the span was the same for all models, and it corresponds to 6 degrees of freedom.

Table 4 presents the estimated percentage increase in respiratory hospital admissions due to an interquartile range increase in air pollutants, using the distributed lag model. For each pollutant, the largest increase in respiratory hospital admissions was observed for the youngest group (from 0–2 years of age). The oldest group (from 14–19 years of age) had the second highest percentage increase in respiratory admissions for PM₁₀ and CO. O₃, SO₂, and NO₂ presented a strong and specific effect on young children, while for the other groups their effects, when positive, were not statistically significant.

Figure 2 shows the behavior of the five pollutants (5-day moving averages of PM₁₀, O₃, CO, and NO₂, and the 4-day moving average of SO₂) when in single-pollutant models and when together in a multipollutant model, with respiratory admissions for all ages as dependent variable. In order to compare equivalent variations in air pollutant-specific exposures, coefficients and standard errors were estimated for a standard deviation increase of each

TABLE 2—Pearson Correlation Coefficients Between Air Pollutants and Weather Variables¹

	PM ₁₀	O ₃	SO ₂	CO	NO ₂	Temp
O ₃	0.16					
SO ₂	0.61	0.17				
CO	0.60	−0.07	0.47			
NO ₂	0.62	0.34	0.54	0.58		
Temp	−0.33	0.11	−0.25	−0.19	−0.19	
Hum	−0.59	−0.19	−0.38	−0.31	−0.37	0.10

¹Temp, minimum temperature; Hum, relative humidity.

TABLE 3—Smoothing Parameters for Covariates Used in the Models¹

	Age groups (years)				
	≤ 2	3–5	6–13	14–19	0–19
Spans					
PM ₁₀ , O ₃ , SO ₂ , and CO models					
Time = 1,795 days	80/1,795	127/1,795	129/1,795	300/1,795	83/1,795
Temperature (°C) ^{2,3}	0.8	0.7	0.5	0.8	0.8
Humidity (%) ³	0.7	0.7	0.6	0.4	0.6
Days of the Week	0.5	0.5	0.5	0.5	0.5
NO ₂ models					
Time = 570 days	92/570	112/570	123/570	365/570	94/570
Temperature (°C) ^{2,3}	0.8	0.8	0.6	0.8	0.5
Humidity (%) ³	0.8	0.8	0.7	0.8	0.6
Days of the week	0.5	0.5	0.5	0.5	0.5

¹An indicator for holidays was also included in all models.

²Minimum temperature of day.

³Two-day mean.

TABLE 4—Percentage Increase and 95% Confidence Interval in Daily Respiratory Hospital Admissions Due to an Interquartile Range Increase in Air Pollutants, Using the Polynomial Distributed Lag Model

Age (years)	Increase (%) and 95(%) CI				
	PM ₁₀ (35 µg/m ³)	O ₃ (46 µg/m ³)	SO ₂ (14µg/m ³)	CO(3 ppm)	NO ₂ (80 µg/m ³)
≤ 2	9.4 (7.9, 10.9)***	1.6 (0.1, 3.0)**	5.9 (4.5, 7.4)***	5.0 (3.3, 6.8)***	9.4 (6.2, 12.6)***
3–5	3.3 (0.2, 6.3)**	−0.0(−3.0, 2.9)	1.6(−1.3, 4.4)	4.9 (1.4, 8.5)**	1.6(−6.4, 9.6)
6–13	1.9(−1.2, 5.0)	0.1(−2.8, 3.0)	0.6(−2.2, 3.5)	1.0 (−2.5, 4.6)	2.3(−5.9, 10.4)
14–19	5.1 (0.3, 9.8)**	0.0(−4.5, 4.6)	1.3(−3.2, 5.8)	11.3 (5.9, 16.8)***	−3.0(−15.7, 9.7)
All ages	7.0 (5.7, 8.2)***	0.8(−7.5, 9.2)	4.5 (3.3, 5.8)***	4.9 (3.5, 6.4)***	6.5 (3.3, 9.7)***

**P* < 0.05.

***P* < 0.01.

****P* < 0.001.

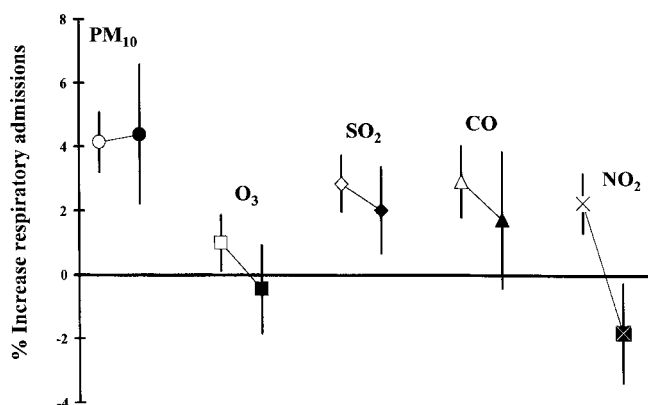


Fig. 2. Percentage increase and 95% confidence intervals in respiratory admissions for all ages together due to a standard deviation increase in 5-day moving average of PM₁₀ (µg/m³), O₃ (µg/m³), CO (ppm), and NO₂ (µg/m³), and 4-day moving average of SO₂ (µg/m³), when they were included in single models (open symbols) and in a multipollutant model (solid symbols).

pollutant. PM₁₀ associations were stable, i.e., resistant to the inclusion of the other pollutants in the model. The O₃, CO, and NO₂ associations became nonsignificant after adjustment for other pollutants. SO₂ was reduced about 30% after the inclusion of other pollutants in the model, but remained positive and statistically significant.

DISCUSSION

This analysis was designed to assess possible differences in the response of children and adolescents to air pollutant exposure. Air pollution was positively associated with respiratory hospital admissions for children and adolescents in São Paulo. The association was estimated using multivariate models, controlling for confounding by seasonality and weather, and adjusting for overdispersion. The magnitude of the estimated effects varied across different age groups and by specific pollutant. Children equal to or younger than 2 years of age

were the most susceptible group, and associations were observed with each pollutant. Older adolescents (from 14–19 years of age) were the second most susceptible group, although only PM_{10} and CO were positively and statistically associated with increases in admissions. When a multipollutant model was used, PM_{10} was resistant to the inclusion of other pollutants, while SO_2 coefficients dropped substantially, CO became nonsignificant, and NO_2 became negative. The results of the multipollutant analysis must be viewed with caution because of the high correlation between pollutants. Adjusting for all pollutants simultaneously could produce apparently nonsensical associations for individual pollutants (e.g., NO_2). However, this multipollutant analysis does indicate the robustness of the PM_{10} associations to maximum adjustment for all other pollutants simultaneously.

The adequate control for autocorrelation demanded specific adjustments for time trend in each age group, as well as in the whole group (Table 3).

We included the most probable and the most commonly used confounders in the association between air pollution and respiratory morbidity: temperature, humidity, and days of the week. The 2-day moving averages for both temperature and humidity gave the best adjustment in the model. The number of holidays and their influence on hospital admissions necessitated inclusion of a specific indicator. Adjusting for holidays increased the association between air pollution and respiratory admissions. Even so, the possibility of incomplete adjustment for confounders always must be considered.

Similar to other authors, we found an enhanced response of infants to air pollutants. This period of life has the greatest physical growth and development, comparable only to the changes in adolescence. In contrast to the adolescent, the infant's immunological system is developing and frequently unable to control adequately the insults suffered by children, resulting in more severe episodes of respiratory diseases when compared to those observed in older children and adolescents. Therefore, infant respiratory infections and other respiratory conditions usually are treated more aggressively and admissions are more frequent than for older children. Conditions associated with pregnancy, such as prematurity, congenital malformations, and low birth weight could contribute to increase the susceptibility of this group of children.

Adolescents older than 13 years of age appeared to be the second most susceptible group to air pollution. Despite similar rates of respiratory diseases, older adolescents were more affected by air pollution than children between 3–13 years of age. Which characteristics of these adolescents make them more susceptible? First, adolescents usually are more exposed to outdoor air pollution. According to research carried out by the State

Data Analysis System Foundation (SEADE),⁴⁰ the percentage of adolescents that both study and work in São Paulo is around 20%. Psychosocial factors and hormonal alterations during adolescence affect physiologic functions,^{41,42} and a decrease in immunological system activity could be expected. The significance of these changes in facilitating respiratory infections is not understood. The initiation of active tobacco smoking may increase the susceptibility of adolescents to air pollution. Adolescent smokers have more respiratory health disorders, including decreases in pulmonary function and increases in respiratory symptoms.⁴³

Analyses taking into account specific respiratory diseases, i.e., infectious diseases and asthma, must be done and must address very important points on the specificity of air pollution effects on respiratory diseases. On the other hand, age-specific susceptibility cannot be explained as merely a consequence of different disease rates. The percentage distribution observed in the oldest age group was very similar to that observed for the group 6–13 years old. However, the response to air pollution exposure was stronger among the adolescents, suggesting that some inherent age-specific characteristics, related to the person and his own environment, must be playing a part. In order to address this issue in the extreme age groups, studies based on individual information are required.

Like many US cities, the main source of pollutants in São Paulo is motor vehicle exhaust. The industrial contribution to PM_{10} , CO, SO_2 , and NO_2 concentrations in the air of São Paulo have decreased substantially since the mid-1980s.³¹ Concentrations of SO_2 well below the air quality standard of $80 \mu g/m^3$ have been observed in São Paulo, and levels of CO were lower than the mean observed in US cities.⁴⁴ A different situation is observed for PM_{10} and NO_2 . Concentrations of inhalable particles (PM_{10}) in São Paulo are high, with annual means frequently above the air quality standard of $50 \mu g/m^3$. Levels of NO_2 have decreased in São Paulo in the last decade, but they are still above the standard of $100 \mu g/m^3$ for the annual mean. In comparison, NO_2 concentrations as high as in São Paulo are rare in the United States.

There are at least two distinct air pollution effect patterns according to age group. SO_2 , NO_2 , and to a lesser extent O_3 presented a higher contribution to respiratory admissions in the youngest group, decreasing in magnitude as children aged. On the other hand, except for the group of children and adolescents from 6–13 years of age, PM_{10} and CO affected all other age groups in a consistent way. The higher contribution of CO to respiratory admissions in the oldest group was unexpected. Particles have been suggested to be facilitators and propagators of infectious diseases, specifically pneumonia.⁴⁵ On the other hand, the role of CO in causing or contributing to the onset of pneumonia and other res-

piratory infections, either as facilitator or propagator, seems unlikely.

In summary, this study showed a significant association between air pollution and respiratory hospital admissions for children and adolescents in São Paulo, and this effect varied by age and specific air pollutant. Children 2 years and under seem to be the most susceptible group, and adolescents (older than 13 years of age) were also significantly affected by air pollution. Further analysis stratifying respiratory admissions by specific causes could help to clarify the specific role of air pollutants in producing respiratory morbidity in susceptible subjects.

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